

PAPER 1 / QUESTION 105

5 November 2025 / TZ3 (file label) / HL Paper 1A / original question 4 / 8825-6719

- 105.** An object of mass m , moving with a speed v , collides with a stationary object of mass m . The objects stick together.

What is the change in kinetic energy in the collision?

A. Zero

B. $\frac{mv^2}{8}$

C. $\frac{mv^2}{4}$

D. $\frac{3mv^2}{8}$